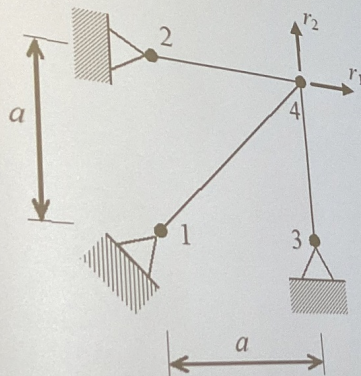
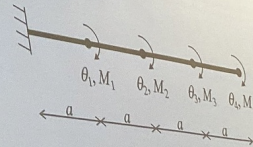


The problems to solve

1. Calculate K_{21} (45%)
Bar stiffness: EA



2. Calculate F_{12} (45%)
Beam stiffness: EI



3. Find the stiffness matrix from the given flexibility matrix given below: (10%)

$$F = \frac{a^3}{6EI} \begin{bmatrix} 2 & 5 \\ 5 & 16 \end{bmatrix}$$

1. Let $r_1 = 1, r_2 = 0$

$$\therefore T_{34} = 0$$

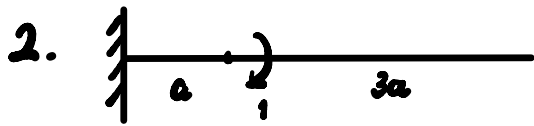
$$T_{14} = \frac{\Delta L}{L} EA = \frac{\frac{\sqrt{2}}{2}}{\sqrt{2}a} EA = \frac{1}{2} \frac{EA}{a}$$

$$T_{24} = \frac{EA}{a}$$

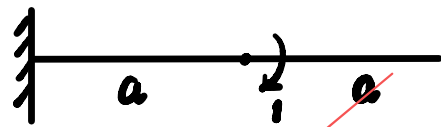
$$\therefore \sum F_v = R_2 - T_{34} - \frac{\sqrt{2}}{2} T_{14} = 0$$

$$\therefore R_2 = \frac{\sqrt{2}}{4} \frac{EA}{a}$$

$$\therefore K_{12} = R_2 = \frac{\sqrt{2}}{4} \frac{EA}{a}$$



$$\therefore M_1 = \begin{cases} 1 & (0 \leq z \leq a) \\ 0 & (a < z \leq 4a) \end{cases}$$



$$\therefore M_2 = \begin{cases} 1 & (0 \leq z \leq 2a) \\ 0 & (2a < z \leq 4a) \end{cases}$$

$$\therefore F_{12} = \int \frac{M_1 \bar{M}_2}{EI} dz = \int_0^a \frac{1}{EI} dz = \frac{a}{EI}$$

3. $K = F^{-1}$

$$= \frac{\text{adj}(F)}{\det(F)}$$

$$= \frac{C^T}{\det(F)}$$

$$= \frac{a^3}{6EI} \frac{\begin{bmatrix} 16 & -5 \\ -5 & 2 \end{bmatrix}}{2 \times 16 - 5 \times 5}$$

$$= \frac{a^3}{6EI} \begin{bmatrix} \frac{16}{7} & -\frac{5}{7} \\ -\frac{5}{7} & \frac{2}{7} \end{bmatrix}$$

$$= \frac{a^3}{42EI} \begin{bmatrix} 16 & -5 \\ -5 & 2 \end{bmatrix}$$